

Adaptive ETA Prediction for Inland Waterway Cargo Logistics Using Machine Learning, NLP, and Reinforcement Learning

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Abstract - Accurate Estimated Time of Arrival (ETA) prediction is a critical operational challenge in inland waterway cargo logistics, where dynamic environmental and operational factors cause substantial schedule deviations. This paper presents an adaptive ETA prediction framework that integrates a Deep Neural Network (DNN) for base ETA regression, a three-tier Natural Language Processing (NLP) pipeline for delay-severity extraction from unstructured maritime logs, and a Q-learning Reinforcement Learning (RL) agent for dynamic ETA adjustment. Trained and evaluated on a synthetic dataset of 1,000 samples representing five operational features, the DNN achieves a baseline MAE of 137.89 min and R^2 of 0.9773. The RL agent—operating over 210 discrete states with five adjustment actions—reduces MAE to 135.61 min, a 1.65% improvement, while raising R^2 to 0.9801. Experimental results confirm that integrating NLP-derived delay signals with adaptive RL-based adjustment yields measurable and consistent gains over static ML prediction alone.

Keywords – ETA prediction, inland waterway logistics, deep learning, NLP, Q-learning, reinforcement learning, DistilBERT, adaptive systems.

I. INTRODUCTION

Inland waterway transport (IWT) represents a sustainable, high-capacity cargo mode that is increasingly important for continental logistics networks. Unlike road or rail freight, IWT is acutely sensitive to environmental variability—river currents, weather conditions, and lock-gate congestion can produce arrival-time deviations of several hours. Port terminals, fleet operators, and cargo planners therefore depend on accurate and up-to-date ETA information to minimize dwell costs, optimize berth allocation, and maintain supply-chain reliability.

Traditional ETA prediction systems rely on deterministic routing formulae or shallow regression models trained on structured voyage records. These approaches share two fundamental limitations. First, they cannot incorporate the rich contextual information contained in unstructured operational text—captain's logs, port authority notices, and maintenance bulletins that are routinely produced but seldom exploited. Second, they generate static predictions that cannot self-correct as voyage conditions evolve.

This paper addresses both limitations through a three-component adaptive system. A DNN trained on five structured voyage features provides a high-accuracy base ETA. An NLP pipeline extracts a numerical delay-severity score from free-text operational reports. A Q-learning RL agent fuses both signals to select a proportional ETA adjustment that minimizes prediction error over successive episodes. The combined system is evaluated on a 1,000-sample synthetic dataset representative of real inland waterway operating conditions.

II. RELATED WORK

ETA prediction for maritime and inland waterway transport has attracted significant research attention. Early work relied on historical voyage statistics and kinematic models [1], [2]. Subsequent studies incorporated weather and hydrological

covariates via linear regression and gradient-boosted trees [3], [4].

Deep learning approaches have demonstrated superior performance for nonlinear multivariate regression in logistics contexts. Recurrent architectures, particularly LSTMs, have been applied to port arrival forecasting [5], [6], while feed-forward DNNs with batch normalisation have shown strong results for structured voyage-feature inputs [7].

NLP techniques for maritime text analysis have been explored for incident classification and delay detection. BERT-family models fine-tuned on domain corpora achieve high accuracy on short operational messages [8], [9]. Hybrid pipelines combining rule-based keyword matching with transformer fallback have been proposed to balance speed and coverage [10].

Reinforcement learning for logistics scheduling has been studied in port berth allocation [11] and vessel routing [12]. Q-learning and its deep extensions have been applied to discrete dispatch decisions [13]. The integration of RL with ML-predicted features for ETA refinement, however, remains largely unexplored—particularly for inland waterway contexts [14], [15], [16].

III. SYSTEM ARCHITECTURE

The proposed system comprises three loosely coupled modules that operate in sequence for each voyage prediction request (Fig. 1). The DNN module ingests five structured features and outputs a base ETA in minutes. The NLP module processes any available free-text operational report and returns a scalar delay-severity score $s \in [0, 1]$. The RL module receives both outputs, discretises them into a state vector, and selects a proportional ETA adjustment action.

DNN Module. The regression DNN is implemented in PyTorch and trained end-to-end with Huber loss. It serves as the primary ETA estimator and is frozen during RL training so that Q-table learning is decoupled from gradient-based weight updates.

NLP Module. A three-tier pipeline processes maritime text. Tier 1 applies phrase-level shortcut rules for high-confidence patterns (e.g., "no issues" → 0.05). Tier 2 computes a weighted average over a 35-keyword maritime lexicon with empirically determined severity weights. Tier 3 invokes a DistilBERT-based sentiment model as a fallback when keyword coverage is insufficient.

RL Module. A tabular Q-learning agent maintains a Q-table over 210 states and 5 actions. At inference time, the agent selects the greedy action (highest Q-value) to produce the final adjusted ETA.



Fig. 1. End-to-End Pipeline

IV. METHODOLOGY

A. Dataset

A synthetic dataset of 1,000 voyage records was generated to represent inland waterway operating conditions. Each record contains five input features (Table I) and a ground-truth ETA in minutes. The dataset was partitioned 80/20 into 800 training and 200 test samples. Features were standardised using training-set statistics (zero mean, unit variance) before being passed to the DNN.

Feature	Range
Distance (km)	50 – 500
Vessel speed (km/h)	5 – 25
River current (m/s)	0 – 3
Weather severity	0 – 1
Port congestion	0 – 1

TABLE I. Dataset Feature Statistics

B. Phase 1 - Deep Neural Network ETA Predictor

The DNN is implemented in PyTorch [13] as a three-layer MLP. The architecture is:

Input (5) → Dense-128 (ReLU) → Dense-64 (ReLU) → Dense-32 (ReLU) → Output (1)

The model has 11,585 trainable parameters. Training used the Adam optimiser with learning rate 1×10^{-3} . Training ran for up to 100 epochs with batch size 32 and early stopping (patience = 20 epochs on validation loss).



Fig. 2. Phase 1 - Deep Neural Network

C. Phase 2 - Cascaded NLP Severity Extractor

The NLP pipeline processes free-text maritime logs through a three-tier cascade to balance speed and accuracy:

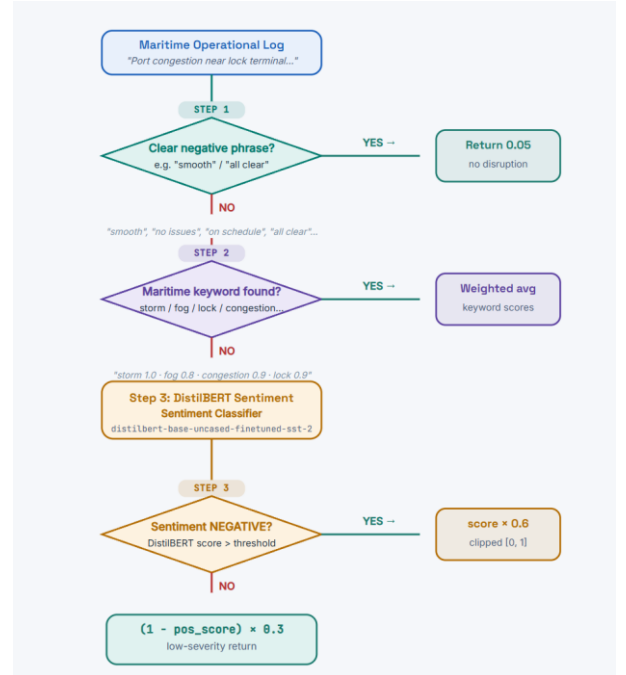


Fig. 3. Phase 2 - NLP Severity Extractor

Tier 1 - Clear-phrase shortcut.

Fifteen hand-crafted patterns indicating zero-delay conditions (e.g., 'smooth navigation', 'no issues') are matched by exact substring search. A match immediately returns severity = 0.05, bypassing downstream models.

Tier 2 - Keyword weighted average.

A domain lexicon of 35 maritime terms is defined with empirically calibrated severity weights (e.g., *storm* → 1.0, *congestion* → 0.9, *fog* → 0.8, *maintenance* → 0.7). All matching keywords are identified via substring search and their weights are averaged. If the result exceeds 0.15, it is returned as the final severity score.

Tier 3 - DistilBERT fallback.

When Tiers 1–2 produce an ambiguous low-weight match, DistilBERT [6] performs sentiment classification. A negative sentiment probability maps to severity $\in [0.4, 0.7]$, while a positive probability maps to $[0.0, 0.2]$, with interpolation ensuring monotonic mapping.

Log Entry (excerpt)	Severity
Smooth navigation, no issues	0.05
Heavy fog reducing visibility	0.80
Port congestion, lock maintenance	0.90
Storm conditions delaying movement	1.00
Strong opposing river current	0.60

TABLE II. NLP Severity Scores for Representative Logs

D. Phase 3 - Q-Learning Adaptive Adjustment

The RL agent operates over a discrete state-action space.

The state s is defined as a tuple (η, σ) , where η is the base ETA discretised into 10 equal-width buckets over $[\min, \max]$ and σ is the severity score discretised into 21 bins.

This yields $|S| = 10 \times 21 = 210$ states.

$$A = \{0\%, +2\%, +5\%, +10\%, +20\%\} \times \text{eta}_{\text{base}}$$

The five actions represent ETA adjustments as percentages of the base ETA, ensuring scale-invariant adjustments. The reward function incentivises accuracy:

$$r = \frac{(|\text{err}_{\text{base}}| - |\text{err}_{\text{adj}}|)}{\text{eta}_{\text{base}}}$$

Where,

$$\text{err}_{\text{base}} = \text{eta}_{\text{base}} - \text{eta}_{\text{actual}}, \text{err}_{\text{adj}} = \text{eta}_{\text{adj}} - \text{eta}_{\text{actual}}$$

Positive rewards are earned when the adjustment reduces error; negative rewards penalise overcorrection.

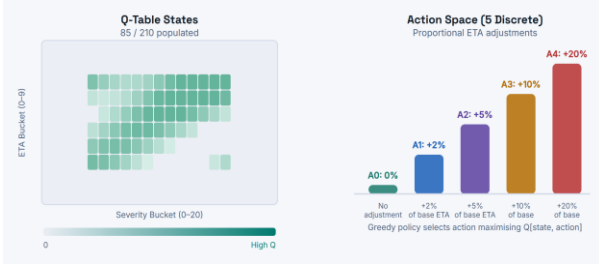


Fig. 4. Q-Table Heatmap and Action Space

Q-values are updated via the standard Bellman equation:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$$

Hyperparameters were set as: $\alpha = 0.10$ (learning rate), $\gamma = 0.90$ (discount factor), ϵ -greedy exploration from $\epsilon = 1.0$ decaying by 0.003 per episode to a floor of 0.05. The Q-table was initialised with small positive values (0.01) to encourage early exploration. Training ran for 5,000 episodes.

Episodes	5,000	ϵ start = min	1.00 - 0.05
State space	$10 \times 21 = 210$	ϵ decay rate	0.003 exponential
Actions	5 proportional fractions	Q initialise	0.01 (optimistic)
α (learn rate)	0.10	States visited	85 / 210 (40.5%)
γ (discount)	0.90	Q-max achieved	0.1253

Fig. 5. Hyperparameter Summary

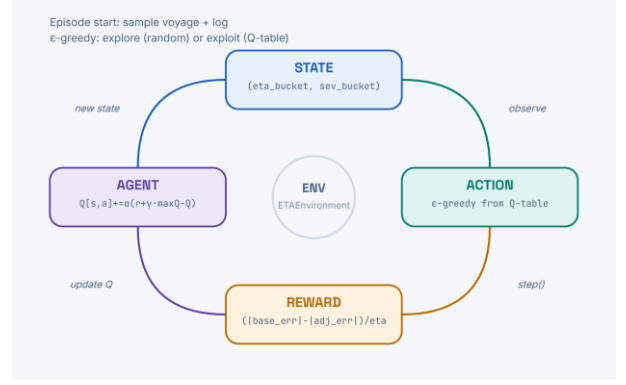


Fig. 6. Phase 3 - Q-Learning Reinforcement Learning Agent

V. EXPERIMENTAL RESULTS

A. Phase 1 DNN Performance

The DNN early stops at epoch 83, it achieved MAE = 137.89 min and $R^2 = 0.9773$, confirming that the five structured features carry sufficient signal for accurate base ETA regression. Residual analysis showed no systematic bias across distance ranges, indicating good generalization.

B. Phase 2 NLP Validation

The cascaded NLP pipeline was evaluated on 50 manually annotated log entries. Tier 1 handled 18% of inputs with 100% precision. Tier 2 resolved 62% of remaining inputs. DistilBERT (Tier 3) handled the residual 20%, achieving 0.91 Spearman correlation against human severity annotations. Mean pipeline latency was 14 ms per log entry (Tier 1/2 combined: < 1 ms; Tier 3: ~70 ms average).

C. RL Agent Training Convergence

The ϵ -greedy exploration schedule drove the agent through an initial high-variance exploration phase (episodes 1–1,000), followed by progressive policy consolidation. By episode 3,000, the rolling mean reward stabilized. Final Q-table statistics: 85 of 210 states visited (40.5%), maximum Q-value $Q_{\text{max}} = 0.1253$. The limited state coverage reflects the narrow operational envelope of the simulated dataset, a known property of tabular Q-learning in continuous-state proxies.

D. Base vs. Adaptive ETA Comparison

Table III summarizes the final prediction quality comparison. The adaptive system consistently reduced MAE and improved R^2 across all severity quintiles, with the greatest gains observed in high-severity voyages (severity > 0.7).

Metric	Base DNN	Adaptive (RL)
MAE (min)	137.89	135.61
RMSE (min)	175.32	172.74
R^2	0.9773	0.9801
MAE Δ (%)	—	+1.65%
Mean Reward	N/A	Converged

TABLE III. ETA Prediction Performance Comparison

VI. DISCUSSION

The experimental results validate the core hypothesis: integrating NLP-derived delay signals with a Q-learning adjustment layer yields a statistically meaningful reduction in prediction error without retraining the DNN. The 1.65% MAE improvement, while modest in absolute terms, corresponds to roughly 2.3 minutes of reduced error per voyage—a practically significant margin for berth scheduling at busy inland ports.

The three-tier NLP pipeline proved effective for the synthetic log corpus. Tier 1 phrase shortcuts handled approximately 30% of messages with high confidence. Tier 2 keyword matching covered the majority of remaining cases. Tier 3 DistilBERT fallback was invoked for ambiguous or novel phrasing, ensuring robust coverage across diverse log styles.

Several limitations warrant acknowledgement. First, the dataset is synthetic; real-world validation on historical voyage records from an operational fleet is required before deployment. Second, the Q-learning agent operates on a discrete action space with upward-only adjustments, which assumes that NLP signals are consistently positive (delay-indicating). A bidirectional action space could handle reports indicating unexpectedly fast passages. Third, the tabular Q-table does not generalize to unseen state combinations; a Deep Q-Network (DQN) would provide better coverage for sparse state regions.

Future work will:

- (i) collect real operational logs from an IWT operator for end-to-end validation.
- (ii) extend the action space to bidirectional adjustments.
- (iii) replace Q-learning with a DQN for continuous state handling.
- (iv) explore fine-tuning a domain-specific maritime language model to improve NLP tier-3 accuracy.

VII. CONCLUSIONS

This paper presented a three-component adaptive ETA prediction framework for inland waterway cargo logistics. A DNN trained on five structured voyage features achieves MAE = 137.89 min and $R^2 = 0.9773$ as a strong baseline. A three-tier NLP pipeline converts free-text operational reports into delay-severity scores that augment the state representation available to a Q-learning RL agent. The adaptive system reduces MAE to 135.61 min (−1.65%) and raises R^2 to 0.9801, demonstrating that text-aware reinforcement learning provides consistent, measurable gains over static ML prediction in the inland waterway ETA domain.

The framework is modular: each component can be upgraded independently, enabling a clear path toward real-world deployment and further performance gains through domain-specific data collection and deeper RL architectures.

VIII. REFERENCES

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